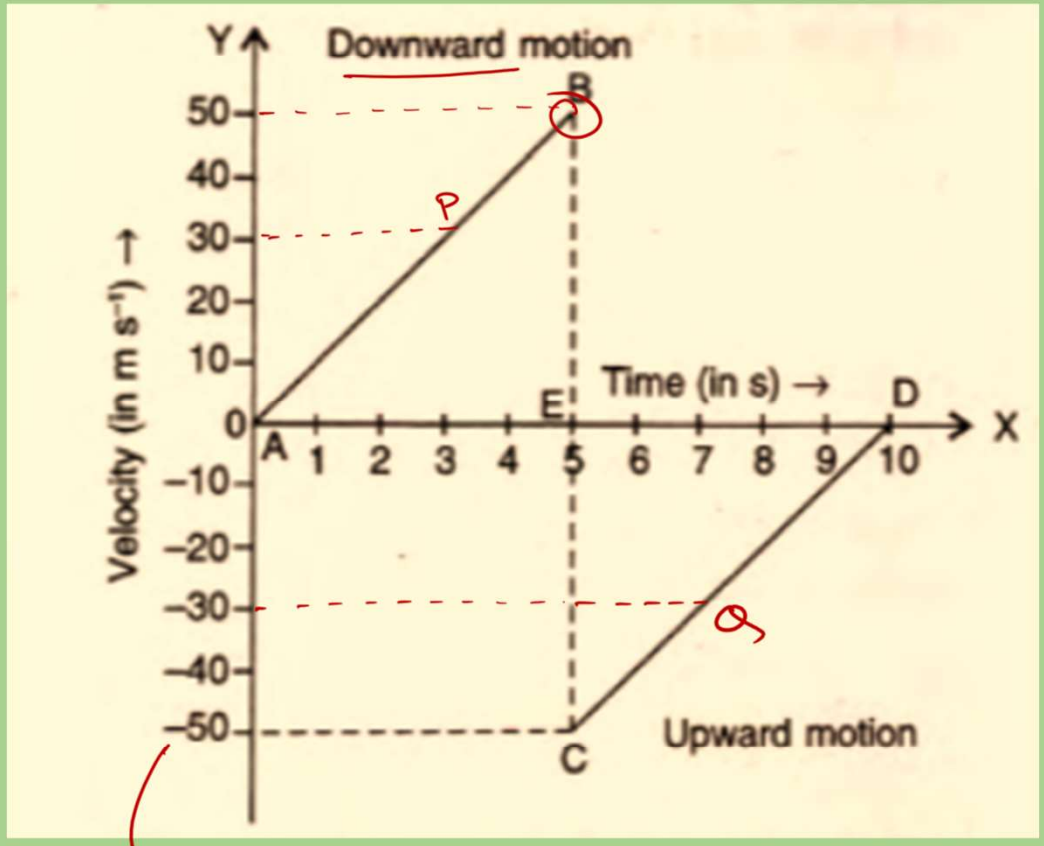
⇒ A ball released from a height and falls freely, after striking the ground it goes vertically upward.

→ If, graph touches X-axis, $y = 0$.

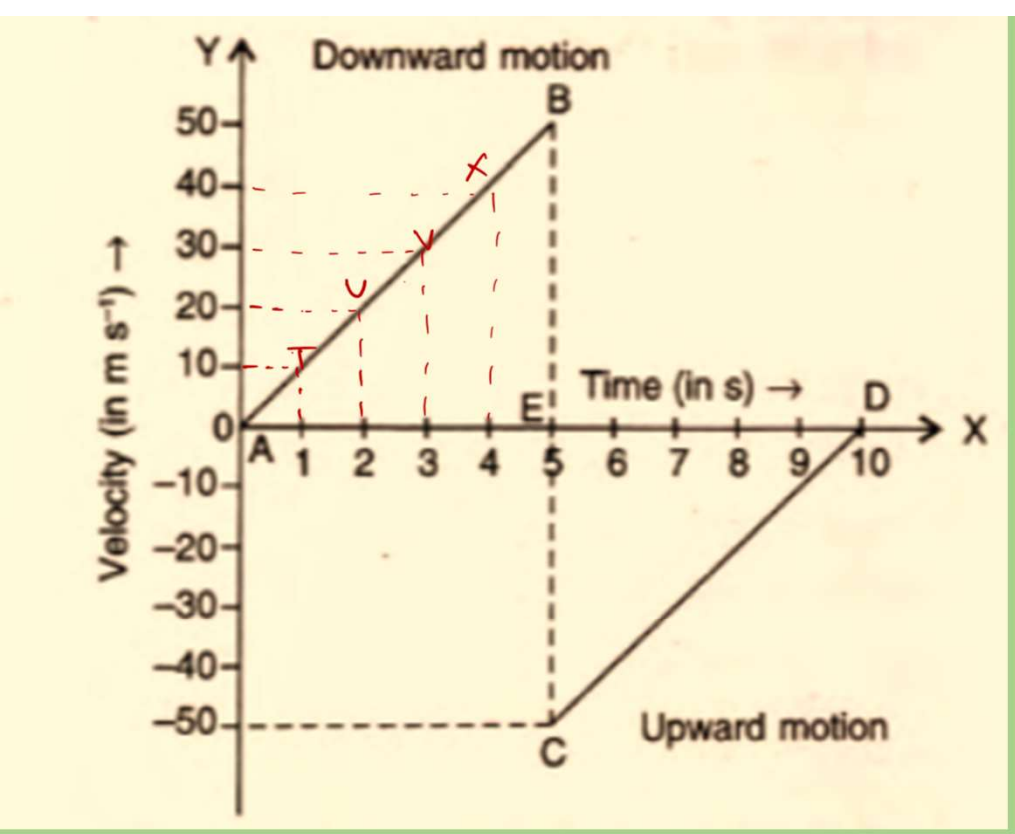
→ If, graph touches Y-axis, $x = 0$



⇒ significance of -ve sign

* velocity at point P = 30 ms⁻¹ due south.
* velocity at point Q = 30 ms⁻¹ due North.

→ The -ve sign is used for direction
↳ above O and below O indicates opposite direction.



Q Acceleration in both upward and downward motion = ?

displacement = ?

distance covered = ?

↳ Acceleration in downward —

$= \frac{50-0}{5-0} = 10\,ms^{-2}$

↳ Acceleration in upward —

$= \frac{0-50}{10-5} = -10\,ms^{-2}$

∴ Retardation $= 10\,ms^{-2}$

↳ Distance covered —

$= \Delta ABE + \Delta CDE$

$= \frac{1}{2} \times 5 \times 50 + \frac{1}{2} \times 5 \times 50$

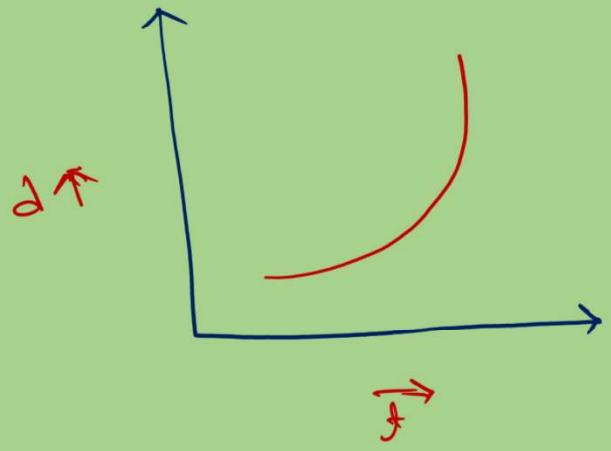
$= 250\,m.$

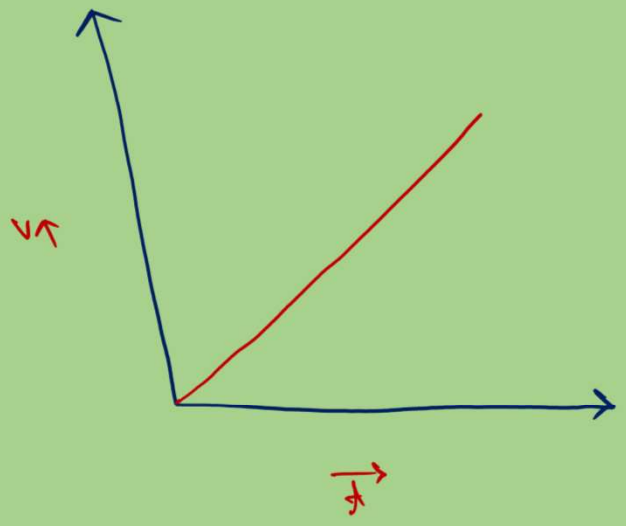
$T = \frac{10-0}{1-0} = 10\,ms^{-2}$

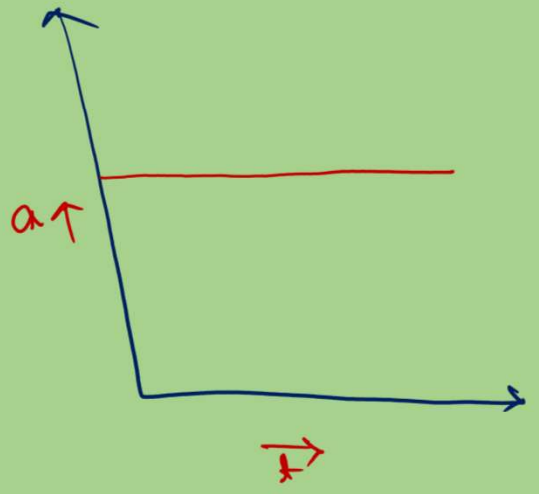
$U = \frac{20-10}{2-1} = 10\,ms^{-2}$

$V = \frac{30-20}{3-2} = 10\,ms^{-2}$

⇒ Motion under gravity (free fall) : Uniformly accelerated motion.







Displacement at —

T = 5 m] = 15m
U = 20 m]
V = 45 m] = 25m
X = 80 m] = 35m.

Velocity at —

T = 10 ms^{-1}] 10 ms^{-1}
U = 20 "]
V = 30 "] 10 "
X = 40 "] 10 ".

Acceleration at —

$T = \frac{10-0}{1-0} = 10\ ms^{-2}$

$U = \frac{20-10}{2-1} = 10\ ms^{-2}$

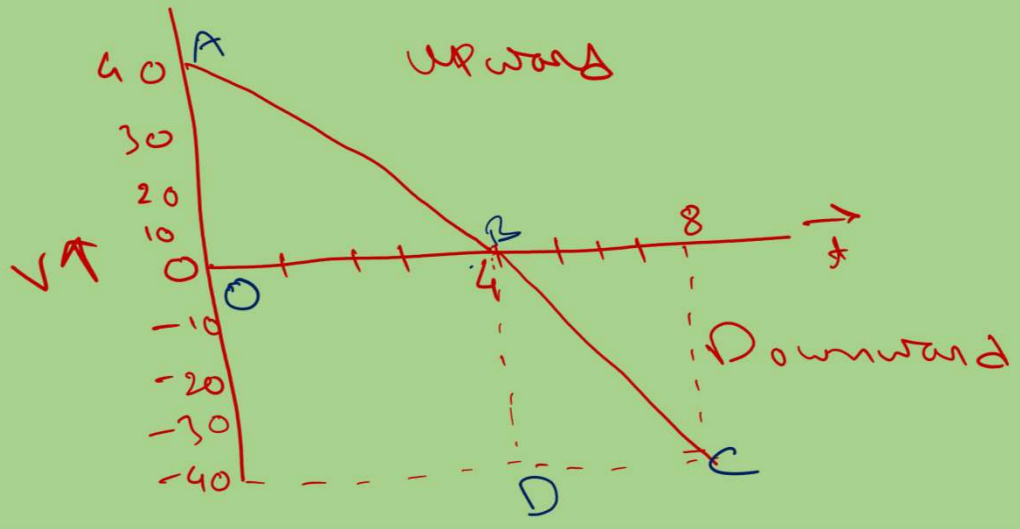
$V = \frac{30-20}{3-2} = 10\ ms^{-2}$

A stone is thrown vertically upwards with an initial velocity of 40 m s^{-1}. Taking $g = 10$ m s^{-2}, draw the velocity–time graph of the motion of stone till it comes back on the ground.

(i) Use graph to find the maximum height reached by the stone.

(ii) What is the net displacement and total distance covered by the stone ?

⇒



→ displacement

i) $= \frac{1}{2} \times 4 \times 40 = 80$ m.

ii) $= \left(\frac{1}{2} \times 4 \times 40\right) + \left\{-\left(\frac{1}{2} \times 4 \times 40\right)\right\}$

$= 80 - 80 = 0$

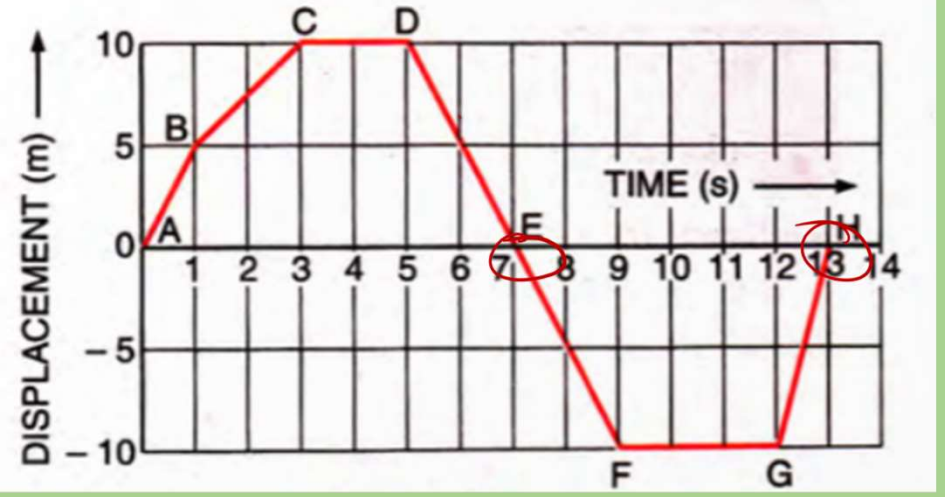
Distance ⇒ $80 + 80 = 160$ m

From the displacement–time graph of a cyclist, given in Fig. 2.39, find :

(i) the average velocity in the first 4 s,

(ii) the displacement from the initial position at the end of 10 s,

(iii) the time after which he reaches the starting point.



7s and 13s.

The velocity–time graph of a moving body is given below in Fig. 2.43.

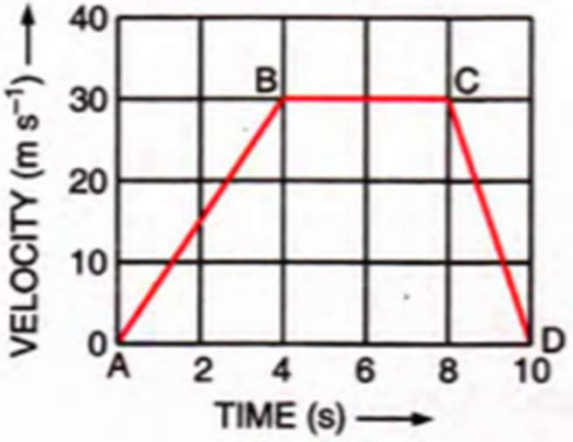


Fig. 2.43

Find :

(i) the acceleration in parts *AB*, *BC* and *CD*.

(ii) displacement in each part *AB*, *BC*, *CD*, and

(iii) total displacement.

i) AB → $\frac{30-0}{4-0} = 7.5\ ms^{-2}$

BC → $\frac{30-30}{8-4} = \frac{0}{4} = 0$

CD → $\frac{0-30}{10-8} = \frac{-30}{2} = -15\ ms^{-2}$

ii) $AB = \frac{1}{2} \times 4 \times 30 = 60m$

$BC = 4 \times 30 = 120m$

$CD = \frac{1}{2} \times 2 \times 30 = 30m$

iii) 210m